

LEDGER · 2026-09-04 · BRAIN-COMPUTER INTERFACES / MATERIALS SCIENCE

Silver-Nanowire / Polyvinyl Alcohol / Melamine Sponge (AgPHMS) Electro-Ionic Semi-Dry Electrodes via Template-Directed Freeze-Thaw Polymerization

CLEARED

This concept cleared all seven gates and every voiding sub-test.

Incumbent benchmark	Sintered Ag/AgCl wet disc electrode
Entry application	wearable, long-term non-clinical BCI research and development sector
Measured advantage	>5.0x increase in duration of stable, ultra-low impedance (<15 kΩ) continuous signal acquisition (>10 hours vs. ~1 to 2 hours for sintered Ag/AgCl wet electrode with gel)
Time to first revenue	4 months
Gross margin at parity	99.4%
Startup CapEx	\$42,000
Payback from first sale	0.1 months

